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<https://doi.org/10.1057/s41599-024-03955-w>

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Computational hermeneutics of emotion: a comparative study of emotional landscapes in the Dostoevsky's novel "Crime and Punishment"

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This study employs computational hermeneutics to explore emotional representations in Fyodor Dostoevsky's *Crime and Punishment*. Using advanced Natural Language Processing (NLP) techniques, including sentiment analysis powered by the BERT model and other tools, we examine the emotional and existential aspects embedded in the narrative. Our approach combines computational tools with traditional hermeneutic methods to deconstruct the emotional journey of the protagonist, Rodion Raskolnikov. We quantitatively analyze his emotional evolution before and after the pivotal crime, mapping his psychological transition from detachment to engagement. The study focuses on four key emotional dimensions: Attitude, Introspection, Sensitivity, and Temper. By transforming qualitative textual data into a quantitative framework, we provide a novel illustration of Raskolnikov's psychological progression from a state of depersonalization to full emotional contact. This computational analysis is contextualized within a broader philosophical interpretation, drawing on concepts such as Stoic *apatheia* and Heideggerian *Sorge*. Our research serves a dual purpose: it enhances our understanding of Dostoevsky's work while demonstrating the potential of computational methods in literary analysis. By bridging digital humanities and traditional literary criticism, this study contributes to the growing field of digital hermeneutics and opens new avenues for interdisciplinary research in literature, philosophy, and computational linguistics.

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Introduction

Over the past few years, the use of Natural Language Processing (NLP) in the analysis of literary works has revealed new opportunities in the study of plot and character transformations. This research uses the following tools of NLP: sentiment analysis and emotion recognition algorithms to analyze the representation of emotions in the novel *Crime and Punishment* by Fyodor Dostoevsky. This paper explores the employment of these computational techniques in order to investigate the emotional contributions of Dostoevsky's works. This allows us to provide an objective and expansive analysis of the author's works according to traditional literary interpretations (Martinsen, 2024). The main hypothesis of this research is that the character of the novel, Rodion Raskolnikov, goes through the emotional arc of disengagement and then engagement, paralleling his moral and philosophical transformation in the novel.

We employ a combination of SenticNet (Cambria et al., 2022) for sentiment analysis and Bidirectional Encoder Representations from Transformers (BERT) (Devlin et al., 2019) for contextual word embeddings to deconstruct the protagonist's primary waves of feelings. This approach allows us to quantitatively track how Raskolnikov's emotions evolve during and after his philosophical insights and actions. By employing these advanced NLP techniques, we aim to present a new illustration of the main character's psychological drift from depersonalization to full emotional contact.

Our methodology integrates computational tools with traditional hermeneutic instruments; this integration serves a dual purpose: enhancing both our understanding of Dostoevsky's work and demonstrating the potential of computational support for literary interpretations. Furthermore, this study contributes to the growing field of digital hermeneutics by showcasing how NLP can provide empirical evidence for longstanding literary and philosophical interpretations.

During the last years, thanks to computational techniques, such as (Auddy et al., 2024; Bonasera, 2023; Feucht et al., 2024; Makinen et al., 2023), many scholars have attempted to go beyond the extraction of the most obvious and easily identifiable meanings to grasp the more profound and easily accessible layers of a text. Emotion extraction and analysis is one of the fields that has greatly benefited from computational approaches, enabling researchers to provide empirical tools for analyzing the emotional charge contained in texts (Klinger et al., 2020).

Emotions are not easy to identify and, nonetheless, they express channels for meaning-making and understanding, enriching therefore the hermeneutic process (Ellis, 2015). When hermeneutic interpretation is extended to include emotions, it fosters a multi-level approach that takes into account how emotional states and transitions contribute to the construction and interpretation of meaning. Rather than being isolated phenomena, emotions are connected with the moral, and psychological dimensions of a text (Ellis, 2015, 2018) thus becoming essential keys to expressing the full interpretive potential of a text.

The paper is structured as follows: First, we present a review of related works in the field of computational literary analysis. Next, we detail our methodology, explaining the specific NLP techniques and tools used. We then present our results, offering both a philosophical interpretation and an analysis of the empirical/computational data. Finally, we discuss the implications of our study for future research in digital humanities and literary analysis.

Related Works

Although there is an evident body of research on computational emotion recognition in literature (Kim and Klinger, 2019; Pavlopoulos et al., 2022; Reagan et al., 2016) studies specifically

aimed at Dostoevsky's works remain limited (Kim and Klinger, 2019). However, the emotional depth in Dostoevsky's narratives presents a rich ground for exploration in this domain (Andersen, 2016).

However, it cannot be denied that the emotional and psychological complexities in Fyodor Dostoevsky's novels have fascinated scholars and readers alike, drawing attention to the intricate intertwining of narrative, emotion, and existential struggle (Sharma, 2017). Thus, for example, the work of (Ali, 2023) points out that Raskolnikov's fragmented psyche serves as a microcosm of broader social tensions, in line with the goal of our study of emotional dynamics. Moreover, (Vanden Auweele, 2016) argue that the emotional landscapes in Dostoevsky's novels are deeply rooted in existential struggles, often manifesting as a clash between rationality and faith.

While traditional approaches offer rich insights, recent works such as (Picca and Richard, 2023), (Kim and Klinger, 2019), (Klinger et al., 2020) and (Rebora, 2023) suggest the potential of computational methods to quantify and analyze emotional complexities.

Methodology

Our study employs a two-stage methodology to analyze emotions in Dostoevsky's novel using SenticNet (Cambria et al., 2022). This text was specifically selected due to its rich emotional representation and complex characters, which make them ideal candidates for an in-depth computational and hermeneutic analysis of emotional dynamics. Moreover, *Crime and Punishment* is considered seminal in Dostoevsky's oeuvre and has been extensively studied in traditional literary scholarship, thereby providing a robust comparative framework for evaluating the insights gained through computational methods.

Preprocessing and emotional lexical expansion. Dialogs within these novels are extracted using the BookNLP tool¹, which identifies and isolates dialogs based on syntactic and semantic cues. Following the extraction, the dialogs are processed for sentiment analysis through a sequence of steps:

1. Tokenization: Each text x is tokenized into a list of words referred to as terms.
2. Initialization: A dictionary scores is initialized to store cumulative scores for four emotional categories: Introspection, Temper, Attitude, and Sensitivity. We reuse the dimensions proposed by Senticnet's Hourglass Model authors which were determined through empirical analysis (Susanto et al., 2020).
3. SenticNet Filtering: A SenticNet DataFrame is filtered to focus on terms appearing in each text x .
4. Emotional Score Calculation: The emotional scores for each category c are calculated by summing the corresponding SenticNet values for each term in t , mathematically represented as:

$$S_c = \sum_{t \in T \cap S} \text{senticnet_filtered}[t, c]$$

Where S_c is the score for emotional category c , t is a term that appears in both the set of tokens T and the SenticNet data S , and $\text{senticnet_filtered}[t, c]$ is the emotional score of term t for category c .

5. BERT Embeddings: To address the issue of insufficient terms in our initial vocabulary, we use a pre-trained BERT-base-uncased model (Devlin et al., 2019). This model generates embeddings $E(t)$ and $E(d)$ for each term t and dialog d respectively.

6. Cosine Similarity: For each dialog d , the cosine similarity is computed between its embedding $E(d)$ and the embeddings of all associated terms t within a specified emotional category:

$$\text{sim}(E_t, E_d) = \frac{E_t \cdot E_d^T}{\|E_t\| \cdot \|E_d\|}$$

7. Score Adjustment: The original emotional score S_t is adjusted based on its similarity to the dialog:

$$S'_t = \frac{S_t \cdot \text{sim}(E_t, E_d)}{\sum \text{sim}(E_t, E_d)}$$

8. Normalization: The scores are normalized to a predefined range $[0, 1]$ using the formula:

$$S'_t = a + \frac{(S_t - \min(S))(b - a)}{\max(S) - \min(S)}$$

where a and b are the bounds of the desired range, and S represents the set of all scores.

The computational processing. The computational processing of Dostoevsky's work is a complex path that is aimed at identifying the 'emotional substrate' of the text. Starting with the analysis of the affective scores for dialogs and lexicon, we try to reveal subtle layers of emotions that Fyodor Dostoevsky managed to hide in his narrative. This first step sets the groundwork for the next steps that we will be undertaking.

To reflect the temporal characteristics of the emotions within the story, the story is divided into the temporal equivalent of the sections. This division helps to trace the changes in the characters' moods and the process of their transformations during the novel.

On top of this temporal structure, we develop the emotional characterization of each character based on the average of the emotions determined through SenticNet. This process provides a numerical measure of each character's emotional state, enabling comparisons and analysis of the character's emotional progress.

To further refine our understanding of character emotions, we employ EmoPlay², we reprocess the extracted dialogs paying much attention to the definition of the emotional features and conditions of the main characters with a special reference to the protagonist Raskolnikov.

While trying to obtain the general picture of the emotions expressed in the novel, we estimate the average values of the emotions for each dialog. This is done by taking an average of the emotional scores of all the constituent words in a dialog; it gives a comprehensive view of the emotional content of specific dialogs. This analysis of the text's narrative enables one to summarise the process of characters' emotional development within the text. When aligning the character dialogs with the dimensions offered by SenticNet, one gets a clear and easily understandable insight into how the characters' sentiments evolve in the course of the story.

This is a pseudo-algorithm of the process³:

Algorithm 1. Context-Sensitive Sentiment Analysis and Emotional Profiling

- 1: **Load Text:**
- 2: Input: Dostoevsky's novel text T
- 3: **Sentiment Analysis:**
- 4: **for** each sentence s in T **do**
- 5: Identify and catalog emotional expressions $E(s)$
- 6: **end for**
- 7: Organize emotions under SenticNet's emotional taxonomy $E(T)$
- 8: **Temporal Division:**
- 9: Divide T into N temporal-equivalent sections $\{T_1, T_2, \dots, T_N\}$

- 10: **for** each section T_i **do**
- 11: Examine temporal changes in emotions $\Delta E(T_i)$
- 12: **end for**
- 13: **Compute Emotional Profiles:**
- 14: **for** each character c **do**
- 15: Calculate the standard deviation $\sigma(E_c)$ of the emotional range
- 16: Gain insights into emotional development and consistency using $\sigma(E_c)$
- 17: **end for**
- 18: **Average Emotional Scores:**
- 19: **for** each character c **do**
- 20: Compute the average emotional score $\mu(E_c)$ from SenticNet
- 21: Build emotional profiles $P(c) = \mu(E_c)$
- 22: Display emotional profiles $P(c)$ graphically
- 23: **end for**
- 24: **Reanalyze Dialogs with EmoPlay:**
- 25: Extract dialogs D as CSV
- 26: Import dialog file D into EmoPlay
- 27: **for** each dialog d in D **do**
- 28: Segment d into words (tokens) $\{w_1, w_2, \dots, w_m\}$
- 29: **for** each token w_i **do**
- 30: Record emotional value $E(w_i)$ from SenticNet
- 31: **if** $E(w_i)$ is not available **then**
- 32: Find synonyms in WordNet and record $E(w_{i_syn})$
- 33: **end if**
- 34: **end for**
- 35: **end for**
- 36: **Compute Composite Emotional Values:**
- 37: **for** each dialog d **do**
- 38: Aggregate emotional values $E(d) = \{E(w_1), E(w_2), \dots, E(w_m)\}$
- 39: Compute composite emotional value $C(d) = \frac{\sum E(w_i)}{m}$
- 40: **end for**
- 41: **Summarize Emotional Profiles:**
- 42: **for** each character c **do**
- 43: Summarize dialogs D_c against SenticNet dimensions
- 44: Compute character evolution of emotions through $\Delta E(D_c)$
- 45: Compare emotional profiles $P(c)$ across narrative segments T_i
- 46: **end for**
- 47: **Output:**
- 48: Generate reports and visualizations of emotional profiles $P(c)$ and development $\Delta E(D_c)$ for each character

This methodology, leveraging the pre-trained BERT-based-uncased model, enables a nuanced and detailed analysis of emotional scores, carefully applied from dialogs to specific terms. This yields probing insight into the emotionally charged layers woven throughout Dostoevsky's dialogic fictions: sensitive shifts in sentiment that complete our emotional grasp of the characters' psychic interiors.

Raskolnikov from indifference to concern: a literary-philosophical interpretation

Before coping with the empirical analysis, we would like to open a methodological parenthesis. Our approach falls within the emerging field of digital hermeneutics. There is growing literature using the term "digital hermeneutics", but this term is still under-defined. We speak of digital hermeneutics in different fields, from digital humanities to computational sociology to critical code studies to philosophy—where one could distinguish between philosophy of mind, political philosophy, media philosophy, etc. The term "digital hermeneutics" has varied interpretations, from code interpretation to the proto-interpretive capabilities of certain

algorithms. It is not our intention to enter the details of the debate here. We simply say that we use the terms “hermeneutics”, “understanding”, “interpretation”, and “explication” in the manner of Ricoeur—in particular, (Ricoeur, 1974). Indeed, Ricoeur’s approach has the following advantages:

1. it overcomes the dichotomy proposed by Gadamer between “truth” and “method”⁴. For the latter, the humanities and philosophy in particular do not need method because, if done well, it directly accesses the truth of its object of study. Against this tradition, Ricoeur asserts that it is possible “to understand better by explaining more” (Changeux and Ricoeur, 2000, p. 215). This means that quantitative explanations and methods do not diverge from the truth of interpretation and understanding but are the condition of its possibility. When he spoke of methods and explanation, Ricoeur had structural linguistics in mind. If he were alive today, he would probably be talking about digital humanities and computational approaches to literature. For us, this means that one should not choose between distant reading and close reading, but that they should be integrated. In a sense, we can say that it is precisely a process of methodological distancing that allows us to have a deeper understanding of certain aspects of the text and its peculiarities;
2. in the specific case of our analysis, we believe that automated emotion extraction can make a significant contribution to the study of emotions and the role that emotions play in interpretive processes. In this regard, Stanghellini and Rosfort (Stanghellini and Rosfort, 2013) have highlighted the importance of Ricoeur’s hermeneutics for understanding the role of emotions in interpretive processes. For them, the heart of hermeneutics is the human effort to make sense of the world. Making sense is in turn related to our values, but values are themselves deeply related to the affective and emotional dimension. The question of meaning is therefore inescapably linked to the affective origin of values.

This is precisely what we are going to observe in the next two sections, first from a literary-philosophical perspective, and then from a computational one: the meaning that we attribute to ourselves, others, and the world depends directly on the emotions we have and cultivate towards them. Although the list of these emotions is extensive, it is however possible to observe two poles, the one that leads to interpreting the world in the light of indifference and closure, on the one hand, and the one that leads to interest, openness, and, ultimately, the perception of the world as rich in possibilities—where, of course, possibility can be a source both of pleasure and pain.

There has long been an ethical and philosophical interest in Dostoevsky and his characters—see, for instance, (Scanlan, 2002). In this context, we will focus particularly on “Crime and Punishment”, which has also been at the center of some ethical and philosophical considerations. For example, one can mention (Bourgeois, 1980, p. 33), who states that “the fundamental pole of opposition in *Crime and Punishment* is the opposition between good and evil, as that which gives structure to the narrative, uniting the complex of oppositions into a meaningful whole”. As for the novel’s protagonist, Raskolnikov, the same author asserts that “[u]nderlying Raskolnikov’s process of degeneration-regeneration is a fundamental way of understanding man in terms of good and evil, and in terms of freedom actualized in each, capable of prostituting itself by becoming enslaved” (Bourgeois, 1980, p. 35).

In this context as well, we will focus on the character of Raskolnikov. Specifically, we propose to understand Raskolnikov’s process of degeneration-regeneration in terms of emotions and

his relationship with the world and others. Our idea is as follows: Raskolnikov’s degeneration-regeneration corresponds to the tension, at the level of moral and existential emotions, that exists in the character between a general antipathy towards human existence (including his own) and his capacity to love and care (sometimes despite himself) for others. The idea of such emotional tension has been advanced by Furtak (Furtak, 2019, p. 48).

In summary, according to Furtak, Raskolnikov does not want to love or care about anyone around him. He wants to practice the Stoics’ *apatheia*⁵. However, as a human being, he cannot truly practice *apatheia*; he cannot avoid care and therefore emotions. For him, *apatheia* represents freedom, while emotions (emotional attachment) represent the opposite. Raskolnikov’s dilemma reflects a profound tension between the desire for emotional independence and the natural human inclination towards empathy and connection with others. This tension can be interpreted through various philosophical lenses, including the distinction between Stoic rationality and the irreducible complexity of human relations emphasized by more contemporary currents of thought. In this context, *apatheia* becomes an almost inaccessible ideal, which clashes with the reality of our being intrinsically social and emotional. Raskolnikov’s search for absolute freedom from emotion leads him to confront his profound humanity. And it is precisely in this hesitation between a life devoid of emotions for others, for oneself, and for the world, and a life that is interested and emotionally rich because it is anchored in concern, that we can see the moral tension between evil and good.

When we speak of “concern”, we primarily think of Heidegger’s notion of *Sorge*. Heidegger (Heidegger, 2007) resorts to the term *Sorge* to describe the fundamental mode of being of *Dasein*, the term he uses to denote the human being or human presence in the world. *Sorge* reflects the intrinsic commitment and concern of *Dasein* for its being-in-the-world. This concept incorporates various dimensions of human experience, including time, mortality, choice, responsibility, and the relationship with other beings. For Heidegger, *Sorge* manifests how the individual projects toward the future, cares about their being, makes choices, and pursues possibilities, thus highlighting the inseparable intertwining between the individual and the world in which they live. Although Heidegger does not outline an opposite of *Sorge* in simple and direct terms, it can be argued that concepts such as “inauthenticity” (*Uneigentlichkeit*) and “falling” (*Verfallen*) function as modes of existence that contrast with *Dasein*’s authentic being, which is deeply rooted in *Sorge*. Both these concepts refer to the process through which the individual becomes absorbed by the everyday world and its superficial concerns, thus moving away from the authentic consideration of their existence. In short, we could say that any distancing from *Sorge* is a distancing from the authentic being of *Dasein*. And this also applies to *apatheia*, whatever the Stoics and Raskolnikov might say. Indeed, the *apatheia* in *Crime and Punishment* seems more akin to what the existentialist Italian novelist Alberto Moravia called “indifference”. The indifference Moravia represents in his first novel (Moravia, 2000) is a lack of feeling, emotion, interest, and concern. But such apathy is very different from that envisioned by the Stoics. For them, *apatheia*⁶ was a means to obtain better knowledge and control of oneself (especially of one’s own passions), while Moravia’s indifference is precisely towards oneself, and one’s own beloved ones.

This is the very point: in Dostoevsky’s perspective, there cannot be an *apatheia* that does not become a form of indifference towards oneself and others, and this indifference is an emotion (or rather, the illusion of a lack of emotions towards the world) that ultimately results in moral and existential evil. Therefore, moral and existential good coincides not with detachment from

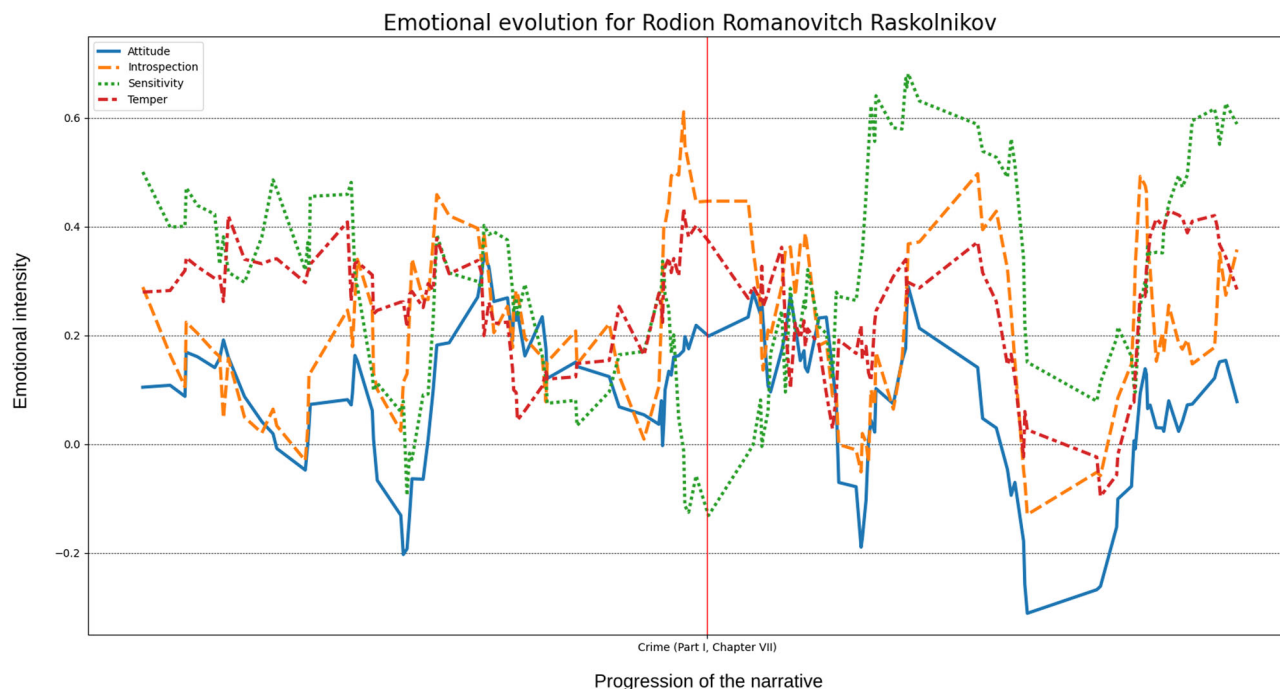


Fig. 1 Emotional evolution of Raskolnikov on the four Senticnet's dimensions. The red line represents the crime.

reality, oneself, others, and the world, but with engagement, and concern, in the sense of care, for these things. Raskolnikov's journey is precisely a growing awareness of this. According to Furtak (Furtak, 2019), his behavior transforms following the crime and the subsequent encounter with Sonya. As an example, consider the dialog in Part 3, Chapter 5⁷ between Raskolnikov, Razumikhin, and Porfiry Petrovich, which revolves around Raskolnikov's article "On Crime". This article argues that certain "extraordinary" individuals have the right to break moral laws. Porfiry summarizes this by stating:

"In his article all men are divided into 'ordinary' and 'extraordinary'. Ordinary men have to live in submission, have no right to transgress the law, because, don't you see, they are ordinary. But extraordinary men have a right to commit any crime and to transgress the law in any way, just because they are extraordinary".

Porfiry uses this discussion to probe Raskolnikov's beliefs and psyche, while Razumikhin defends his friend by suggesting these theories are purely academic. This conversation highlights, among other things, the stark contrast between Raskolnikov's cold detachment and Razumikhin's compassionate, morally grounded nature. Through the act of crime and the suffering that ensues, Raskolnikov gradually comes to revisit his feelings towards the world and the others. Raskolnikov's confession to Sonya (Part 5, chapter 4) is a critical moment that marks the beginning of his moral awakening. Certainly, Raskolnikov continues to resist: "Perhaps I've been unfair to myself, he observed gloomily, pondering, 'perhaps after all I am a man and not a louse and I've been in too great a hurry to condemn myself. I'll make another fight for it'". But Sonya's attitude seems to open a path to repentance:

"They sat side by side, both mournful and dejected, as though they had been cast up by the tempest alone on some deserted shore. He looked at Sonya and felt how great was her love for him, and strange to say he felt it suddenly burdensome and painful to be so loved. Yes, it was a strange and awful sensation! On his way to see Sonya he had felt that all his hopes rested on her; he expected to be rid of at least part of his suffering, and now,

when all her heart turned towards him, he suddenly felt that he was immeasurably unhappier than before".

This process, though not without ambiguity, leads him to open up to the various ways in which he finds himself to be part of the human fabric, a realm where the importance of people, interactions, and things emerges and becomes meaningful to him. Furtak refers to (Noddings, 2003, p. 51) on this point, who states: "I am totally free to reject the impulse to care, but I enslave myself to a particularly unhappy [fate] when I make this choice. As I chop away at the chains that bind me to loved others, asserting my freedom, I move into a wilderness of strangers and loneliness [...]".

The purpose of this paper is not simply to further argue around this thesis-interpretation about *Crime and Punishment*. Instead, our idea is to look for quantitative evidence of it, that is, explanations based on the methods of digital humanities and particularly on the detection of emotions in literature. In short, our idea is that the quantitative analysis of Raskolnikov's emotions can demonstrate that he does indeed change his attitude towards the world, himself, and others throughout the novel, moving from a state of indifference to one of concern.

Raskolnikov from Indifference to Concern: An Interpretation of the Computational Analysis

In our analysis of Dostoevsky's *Crime and Punishment*, we apply digital humanities methodologies to examine the emotional profile of its characters, particularly focusing on the protagonist, Raskolnikov. The digital hermeneutics approach, drawing upon Ricœur's principles, allows us to integrate distant reading with close reading techniques, thereby enriching our understanding of the narrative through both quantitative and qualitative insights.

In Fig. 1, we quantify emotional dimensions based on SenticNet's categories for Raskolnikov. The graph tracks four emotional metrics: Attitude, Introspection, Sensitivity, and Temper throughout the narrative. The vertical red line in the middle marks the pivotal event of the crime.

Before the crime, Raskolnikov's emotional metrics display fluctuations but within a relatively restrained range, suggesting a

certain level of emotional containment or *apatheia*. Indeed, before the crime, Raskolnikov is depicted as detached and isolated, exemplified by his reluctance to engage with others, such as when he abruptly ends a conversation with Marmeladov and feels a loathing at the mention of his university studies, which were interrupted due to poverty. The moments leading up to the crime seem characterized by internal tension, as indicated by the drop in Sensitivity, representing a withdrawal from the emotional engagement with others and the world—a critical juncture that echoes Heidegger’s concept of “thrownness” (*Geworfenheit*) into a situation. Post-crime, there’s a notable increase in the variability and intensity of all emotional dimensions. This heightened fluctuation can be interpreted as a shift from *apatheia* to *Sorge*—Raskolnikov’s profound concern and care for his own being and his relationships with others and the world. The volatility of emotions after the crime reflects the inner turmoil and existential dread that follows his action, aligning with Heidegger’s philosophy where *Sorge* becomes the fundamental way through which one is thrown into the world and engages with it. For instance, after the crime, Raskolnikov’s interactions with Sonya illustrate a shift towards *Sorge*. He questions the morality of his actions in hypothetical discussions with her and, despite trying to justify the murder, ultimately reveals his crimes to her, indicating a deeper need for connection and absolution.

The shifts in Attitude, Introspection, and Temper especially after the crime, are indicative of a person grappling with the consequences of his actions, reflecting an increased preoccupation with moral and existential questions. This analysis reinforces our argument that Raskolnikov’s character evolves from detachment to a state of profound engagement.

In Fig. 2, we provide a look at the changes in Raskolnikov’s emotional intensity throughout the novel. The average emotional intensity calculated as 0.595 before the crime and 0.609 after shows a slight increase. While this difference is minimal, it suggests a shift in Raskolnikov’s emotional state. For instance, before the crime, Raskolnikov exhibits emotional detachment and apathy, as seen in his interactions with Marmeladov and his disdain

for his interrupted studies. After the crime, his heightened emotional state is evident in his interactions with Sonya and his paranoid behavior. These qualitative examples highlight the emotional intensity that numerical averages might not fully capture. To better understand the variability in Raskolnikov’s emotional intensity, we also calculated the standard deviations: 0.2313 before the crime and 0.2356 after. These measures indicate greater emotional fluctuations post-crime, supporting the qualitative observations of increased emotional turmoil. While the quantitative analysis shows a slight increase in emotional intensity, this finding should be interpreted with caution. Further textual analysis is necessary to substantiate these results and provide deeper insights into Raskolnikov’s emotional transformation.

From a hermeneutical point of view, before the crime, Raskolnikov often exhibits apathetic and withdrawn behavior, indicative of low emotional intensity. For example, his inability to find joy in life and his seclusion from society demonstrate this lower intensity. He is described as crushed by poverty, but with no clear direction, wandering the streets of St. Petersburg in a state of despondency. After the crime, Raskolnikov’s mental state is far more turbulent. The increase in emotional intensity can be seen through his panic and paranoia, his fluctuating attitudes towards others, and his philosophical struggles with his actions. The psychological turmoil and the bouts of illness he experiences after the crime demonstrate a higher emotional intensity. His conflicting feelings of pride and revulsion about his murderous act, as well as his eventual confession and the complex relationships with characters such as Sonya and Svidrigailov, are testaments to this heightened emotional intensity.

We now present a box plot representing the emotional distribution of each character on the four Senticnet’s dimensions. Our methodology employs sentiment analysis algorithms that interpret linguistic cues related to emotional expressions in the text, quantifying them into measurable indicators.

Figure 3 compares the profiles of various characters from “Crime and Punishment” along four dimensions derived from

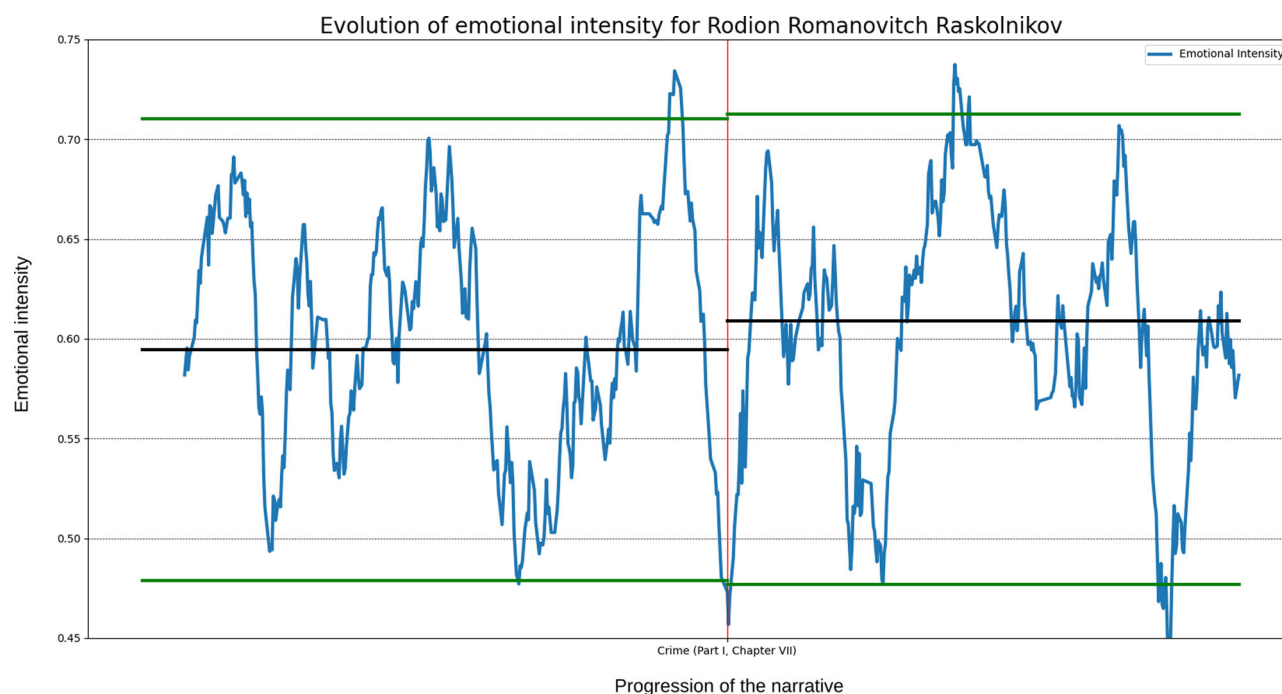


Fig. 2 Raskolnikov’s emotional intensity evolution through the text. The red line represents the crime. The black lines represent the average value before and after the crime, respectively. The green lines represent the standard deviation before and after the crime, respectively.

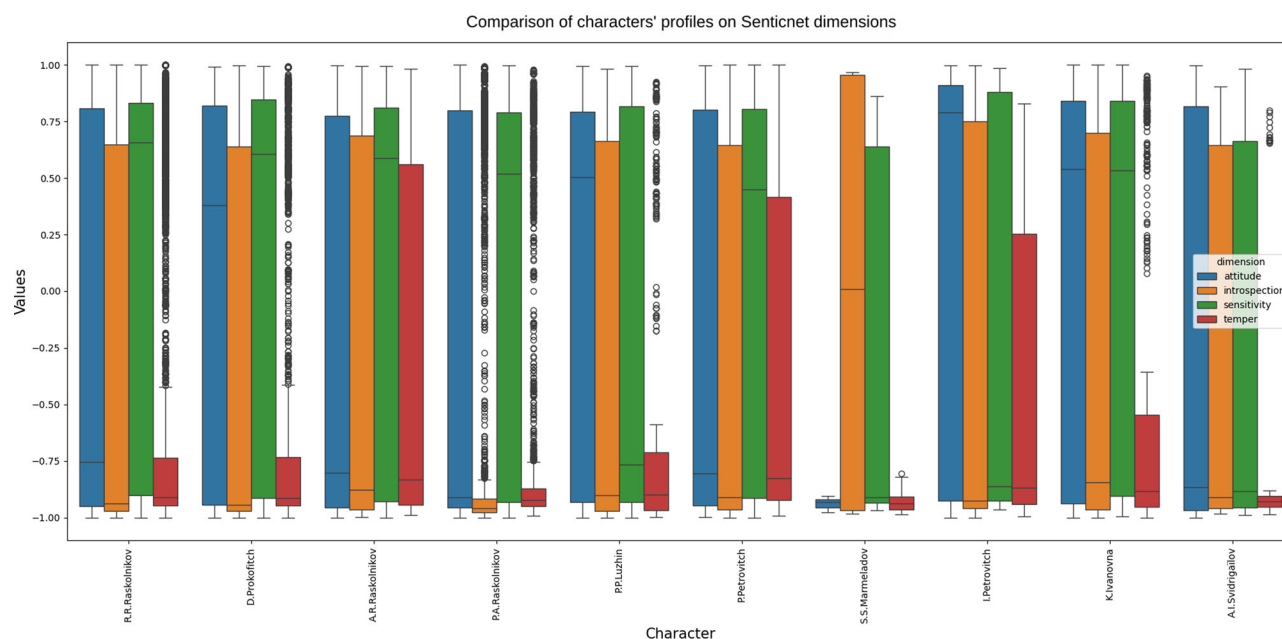


Fig. 3 Comparative analysis of sentiment dimensions in characters from Dostoevsky's "Crime and Punishment". The bar chart represents the average values for five dimensions: attitude (blue), introspection (orange), sensitivity (green), temper (red), and other key dimensions across different characters. The y-axis shows the values on a normalized scale from -1 to 1, where positive values indicate positive expressions of the respective dimension and negative values indicate negative expressions. Error bars represent the standard deviation across sampled texts for each character, and outliers are indicated as individual points. Characters include R. Raskolnikov, D. Prokofitch, P.A. Raskolnikov, P.P. Luzhin, and others, allowing a nuanced exploration of their emotional and psychological complexity.

SenticNet (See Fig. 4): Attitude, Temper, Introspection and Sensitivity. Each character is represented by four bars, one for each Senticnet dimension.

The box plot illustrates the emotional profiles of various characters across four SenticNet dimensions: Attitude, Introspection, Sensitivity, and Temper. The whiskers represent each character's emotional variability, which provide a deeper understanding of the range and fluctuations in emotional responses, rather than relying solely on median values. This method captures the nuances of how emotions manifest in different contexts, offering a more detailed view of each character's emotional landscape.

Raskolnikov, the protagonist, exhibits substantial variability in the "Temper" dimension, indicative of extreme distress or negativity. His emotional profile reveals a character grappling with profound inner conflict and instability, characterized by periods of rationalization punctuated by intense outbursts of passion, anger, or despair. This is reflected in the wide range of values and the significant standard deviation observed in his "Temper" scores.

The plot further distinguishes between different emotional expressions among the characters, forming distinct clusters. One cluster, which aligns closely with Raskolnikov's emotional range, demonstrates higher variability and intensity. This group includes characters such as D. Prokofitch and A. Raskolnikov, whose emotional profiles suggest a shared experience of distress and negativity. Their proximity to Raskolnikov in the narrative may indicate a closer relational or psychological connection, mirroring his inner turmoil and existential struggles.

Conversely, another group exhibits a temperament leaning towards calmness and positivity, indicative of a more stable or detached emotional state. Characters like P. Luzhin and K. Ivanova show less emotional variability and lower intensity, reflecting their more distant or oppositional stance to Raskolnikov's internal conflict. Their emotional profiles suggest a lesser

degree of engagement with the extremes of Raskolnikov's emotional spectrum, possibly highlighting their external or contrasting roles within the narrative.

Thanks to such insights, we propose an interpretation of Raskolnikov's emotional profile. Intellectually, he espouses Stoic *apatheia*, striving for emotional detachment from worldly engagements. However, the evolution of the "Temper" dimension in Fig. 1 contradicts this intellectual stance, revealing that, in practice, Raskolnikov remains deeply affected by and engaged with his surroundings. His frequent shifts in temper reflect an individual inescapably tied to the very emotions he aims to suppress. Our findings illustrate a key narrative tension: the intellectual aspiration towards Stoic dispassion contrasted against the natural human propensity for emotional engagement. Dostoevsky's character does not live in emotional isolation; rather, he is an active participant in the human condition, embroiled in a web of social and emotional connections. The emotional analysis underscores a fundamental humanistic theme of the novel: the struggle between the ideal of detached rationality and the undeniable pull of empathy and moral concern. The contrast between Raskolnikov's intellectual justification of the crime in his article "On the Crime" and the genuine emotions he experiences during his confession to Sonya highlights this contradiction clearly.

Conclusion

The paper employed methods like automated emotion extraction and clustering to examine the emotional territories in Fyodor Dostoevsky's *Crime and Punishment*, with regards to characterization and the rationale for action by Rodion Romanovich Raskolnikov. Combining the more conventional close reading with the use of digital humanities provided an even deeper perspective into the social, emotional, and psychological processes at work within the Dostoevsky story.

The study showed that Raskolnikov's emotional arc changes dramatically as the novel progresses. From the beginning,

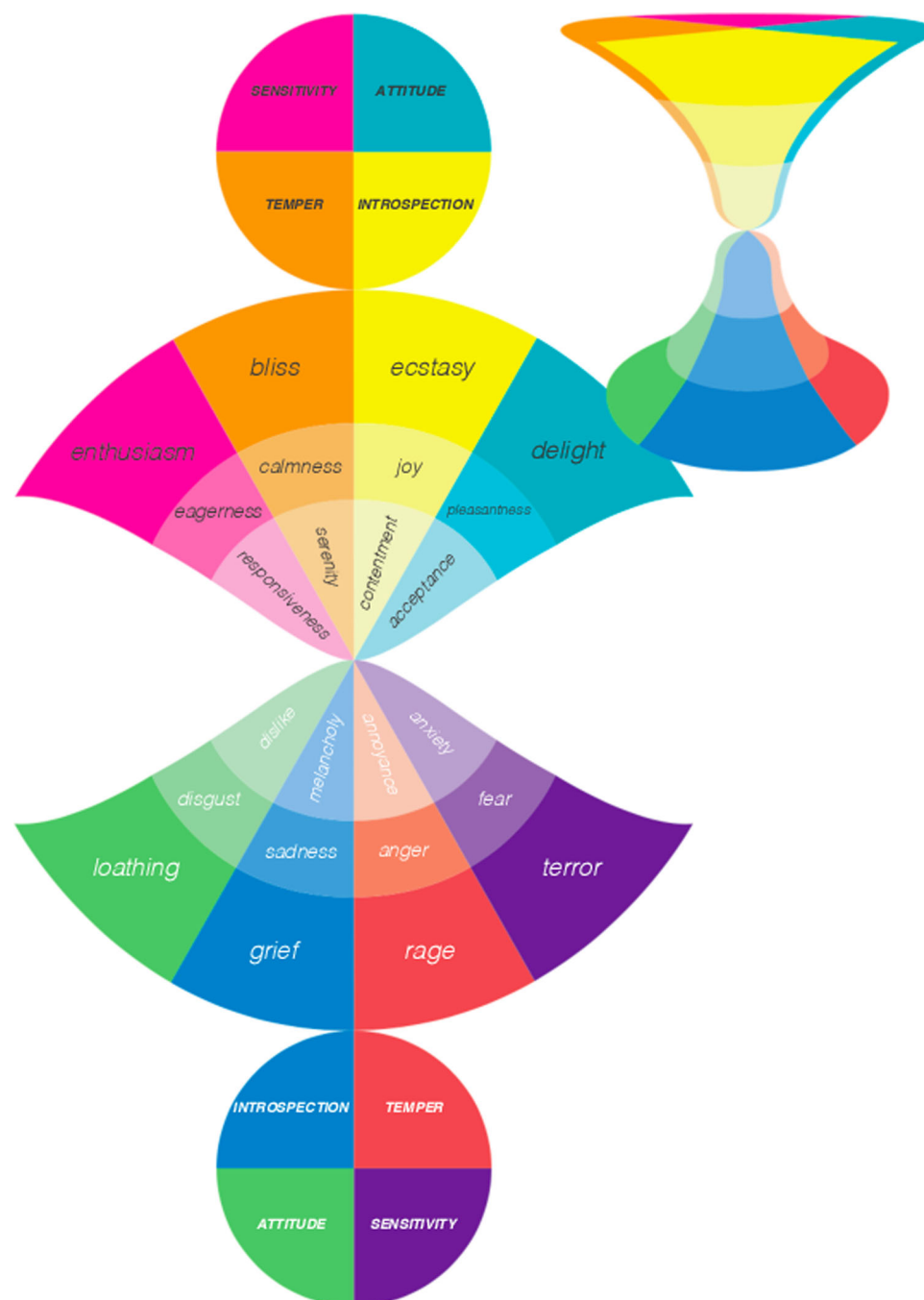


Fig. 4 Sentiment and emotion mapping based on SenticNet dimensions. This figure represents the relationships between primary sentiment dimensions (sensitivity, attitude, introspection, temper) and associated emotions. The circular segments illustrate how the four core dimensions—sensitivity, attitude, introspection, and temper—correspond to specific emotional states. Positive emotions, such as ecstasy, joy, and delight, are mapped in the top half of the diagram, transitioning through layers of related emotions like enthusiasm, eagerness, and calmness. The bottom half depicts negative emotions, such as grief, rage, and terror, with further subdivisions into nuances like melancholy, anxiety, and sadness. The visual depiction resembles a funnel, demonstrating how broad dimensions break down into more specific emotions, capturing the complexity of human emotional experience (Susanto et al., 2020).

Raskolnikov's emotions were shaped by a Stoic indifference, manifest in his desire for apatheia; however, his emotional trajectory changes significantly after the murder act, which is to be decisive. The use of tools such as SenticNet and EmoPlay enabled us to quantitatively demonstrate changes in four emotional dimensions: Attitude, Introspection, Sensitivity, and Temperament. After crime, such shifts were especially significant when the latter depicted passion in a higher degree and storm—another qualitative change that indicates the amplification of his interest in existential and ethical issues.

Therefore, the findings would substantiate the hypothesis that the process of Raskolnikov changing from indifferent to concerned is not, in this case, some philosophically or ethically motivated story but is brought to very real life through his moods. This paradigm shift unveils a focal tension in the major theme: the personal journey of the man who is between the capacity to reason and the innate propensity for compassion and concern for the other.

This is another example of how digital hermeneutics can add back to literature—in this case, evidence for what Paul Ricœur

himself thought: In order to come to a better understanding, one needs to add an interpretative approach to an explanatory one. It verified some traditional views and, at the same time, suggested to some extent new interpretations of the emotional basis of the development of the character in a literary text. The findings of this study thus not only contribute to the larger field of digital humanities but signal a way in which computational analysis may cross with and enlarge literary hermeneutics to their own mutual advantage. Building on the insights gained from this study, future research could apply similar computational hermeneutics techniques to other complex literary works, such as those of Tolstoy or Kafka, to uncover new dimensions of character development and emotional dynamics. By combining sentiment analysis with other NLP tools, such as topic modeling and network analysis, researchers can further elucidate the interplay between plot structure, character interaction, and emotional evolution, thus paving the way for a more holistic approach to literary analysis.

Data sharing

The datasets generated during and/or analyzed during the current study are available https://github.com/unil-ish/Dostoyevsky_EmoPersonality.git and are fully accessible upon reasonable request.

Received: 30 April 2024; Accepted: 16 October 2024;

Published online: 26 October 2024

Notes

- <https://github.com/booknlp/booknlp>.
- This is done through the use of a specialized tool for emotional analysis which can be found at: <https://anonymous.4open.science/r/EmoPlay-7CFB/>.
- The code is available at the address: https://github.com/unil-ish/Dostoyevsky_EmoPersonality.git.
- "The question is to what extent [Gadamer's] work deserves to be called Truth AND Method, and whether it ought not to be entitled instead Truth OR Method" (Ricoeur, 1991, p. 71).
- Stoic *apatheia* refers to a state of inner peace achieved through the absence of disturbing passions, such as desire, fear, pain, and excessive pleasure. The Stoics believed that by attaining *apatheia* one could live in harmony with reason and nature, accepting external events without being emotionally perturbed by them. The goal was to promote practical wisdom and virtue that would allow an individual to maintain inner tranquility in the face of life's adversities.
- At the heart of the novel are two siblings from a bourgeois family, Carla and Michele Ardengo, who are incapable of experiencing true emotions, caught in a state of boredom and indifference amidst the social and economic decline of their family. Their mother, Mariagrazia, a widow, leads a habitual life clinging to the moral clichés of the bourgeoisie, remaining largely unaware of their plight. In Moravia's view, boredom and indifference serve as mechanisms for detaching and isolating oneself from reality and the world. Carla and Michele's inability to feel genuine emotions reflects a broader existential crisis, where indifference becomes a shield against the harsh realities they face.
- All citations from *Crime and Punishment* are taken from Constance Garnett's translation, which is available online through Project Gutenberg (<https://www.gutenberg.org/ebooks/2554>). As this version is in HTML format, no page numbers are provided.

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Author contributions

Davide Picca, Antonin Schnyder, Alberto Romele contributed equally to this work.

Competing interests

One of the authors was a member of the Editorial Board of this journal at the time of acceptance for publication. The manuscript was assessed in line with the journal's standard editorial processes, including its policy on competing interests.

Ethical approval

This article does not contain any studies with human participants performed by any of the authors.

Informed consent

This article does not contain any studies with human participants performed by any of the authors.

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